

16	Two polarising filters are placed next to each other so that their planes are parallel. The first polarising filter has its transmission axis at an angle of 50° to the vertical. The second polarising filter has its transmission axis at an angle of 20° to the vertical. The angle between the transmission axes of the two polarising filters is 30°. □ □ A beam of vertically polarised light of intensity 8.0 W m^{-2} is incident normally on the first polarising filter. What is the intensity of the light that is transmitted from the second polarising filter?							
	A	zero	B	2.5 W m^{-2}	C	2.9 W m^{-2}	D	6.0 W m^{-2}

Answer: B

Light passes through the first polarising filter.

Since the light is initially vertically polarised, the angle with the first filter's axis is 50° .

Using Malus's Law:

$$I_1 = I_0 \cos^2(\theta) = 8.0 \times \cos^2(50^\circ) = 8.0 \times (0.6428)^2 = 8.0 \times 0.4132 = 3.31 \text{ W m}^{-2}$$

Light passes through the second polarising filter.

The angle between the transmission axes of the two filters is 30° .

Again using Malus's Law:

$$I_2 = I_1 \cos^2(30^\circ) = 3.31 \times (0.8660)^2 = 3.31 \times 0.75 = 2.5 \text{ W m}^{-2}$$